

An Energy-Based Composite Score: Joules, Not Heuristics

Turning the competition's abstract score into the physical quantity it actually stands for — energy consumed on real hardware.

Why do this

The competition score is a ratio of modeled *times*. That's a fine proxy, but it hides what those times physically *are*: a GPU drawing power, and an interconnect burning energy to move expert weights. Time is the heuristic; **energy is the real, tangible cost**. A balancing policy that lowers modeled time is, literally, a policy that consumes fewer joules to serve the same tokens.

The good news: the simulator already models the two physical activities. Energy is just **power** × **time**, so we can convert its time model into a joules model with no hand-waving — every constant below is a published hardware figure.

The hardware the simulator is modeling

The simulator's transfer bandwidth constant is $(900 \times 10^9 \text{ bytes/s})$. That is not arbitrary — it is exactly **NVLink 4.0's 900 GB/s**, the GPU-to-GPU interconnect on the **NVIDIA H100 SXM**. So the scoring model implicitly assumes H100-SXM-class nodes, and we can attach that platform's real power figures:

Quantity	Value	Source
GPU peak power (TDP)	700 W	H100 SXM datasheet (NVIDIA)
Interconnect bandwidth	900 GB/s	NVLink 4.0 (matches simulator)
NVLink transfer energy	1.3 pJ/bit	NVIDIA (NVLink vs PCIe efficiency)
HBM3 read energy	0.29 pJ/bit	HBM3 interface measurement (FinFET impl.)
Expert size	88,080,384 bytes	simulator constant (expert_bytes)
Compute-seconds per PAR point	60 s	simulator constant $(\text{balanced_compute_seconds})$

The two energy terms

The simulator's $\text{modeled_time} = \text{compute_time} + \text{transfer_time}$. We give each term a wattage.

Compute energy (the cost of imbalance)

A PAR of X means the slowest device does $X\times$ the balanced compute, and the simulator charges $60\text{ s} \times \text{mean_PAR}$. That time runs on a GPU at roughly its TDP, so:

$$E_{\text{compute}} = 700\text{ W} \times 60\text{ s} \times \text{mean_PAR} = 42,000\text{ J per PAR-point, per window}$$

One PAR point $\approx$ 42 kJ of compute energy. This is the expensive term.

Transfer energy (the cost of moving an expert)

Moving one expert means reading $\sim 88\text{ MB}$ from HBM and pushing it across NVLink. Two independent methods bracket the cost:

- **Bit-energy method** (most physical): $704,643,072\text{ bits} \times (1.3 + 0.29)\text{ pJ/bit} \approx \mathbf{0.0011\text{ J per expert moved}}$ (link + HBM read).
- **TDP-window method** (most conservative): the move occupies the link for $88,080,384 / 900 \times 10^9 = 97.9\text{ }\mu\text{s}$, during which the node draws $\sim 700\text{ W} \rightarrow \approx \mathbf{0.069\text{ J per expert moved}}$.

The true per-move cost sits between **0.001 and 0.069 J**. We use the conservative 0.069 J so we never *understate* movement.

The exchange rate, in physical energy

This is the punchline, and it's the same asymmetry the team found heuristically — now in joules:

$$\begin{aligned} 1\text{ PAR point} &\approx 42,000\text{ J} \\ 1\text{ expert move} &\approx 0.069\text{ J} \quad (\text{conservative}) \end{aligned}$$

■ **One PAR point is worth roughly 600,000 expert moves, energetically.**

That single fact justifies the entire design philosophy: balance is the overwhelming energy cost; movement is almost free *per move*. Spending energy to cut PAR is always worth it; spending PAR to cut movement (what v2 did) is burning joules.

The composite energy score

Define total modeled energy exactly like modeled time, but in joules:

$$E_total = 700\text{ W} \times 30\text{ s} \times mean_PAR \times n_cases$$
$$+ 0.069\text{ J} \times transmit_amount$$
$$EnergyScore = 100 \times E_baseline / E_total$$

(compute)

(transfer)

(higher = less energy = better)

Applied to the confirmed version data ($n_cases = 25$):

Version	mean PAR	expert moves	Compute energy	Transfer energy	Total	Energy score
v1	1.722	150,000	1,808 kJ	10.3 kJ	1.82 MJ	100.0
v2	1.779	391,000	1,868 kJ	26.8 kJ	1.89 MJ	95.97

Two things fall out, both important:

- The energy score ranks v1 above v2 (100 vs 95.97)** — the *same* ordering as the real grader (110.86 vs 109.37). The energy model isn't a different game; it measures the same thing the competition does, in honest units.
- Compute is 99.4% of the energy; transfer is 0.57%.** The joules confirm, physically, what the time model only implied: PAR is everything, movement is a rounding error.

The v2 failure, told in joules

This is the cleanest way to explain why v2 lost, and it needs no reference to "score points":

- v2's PAR rose by 0.057. Over 25 cases that is $42,000\text{ J} \times 0.057 \times 25 \approx$ **60 kJ of extra compute energy.**
- v2's transit saving was 241,000 fewer moves: $0.069\text{ J} \times 241,000 \approx$ **16.5 kJ saved.**
- Net: v2 wasted ~43 kJ.** It optimized the 0.5% term and paid for it in the 99.5% term.

v2 didn't just lose points — it physically consumed more energy to do the same work.

Making it tangible at production scale

Per short grader run the absolute joules are small (it's a simulation of a few windows). The honest way to make this concrete is to scale to **continuous serving**, where these policies

actually live. A single 8-GPU H100 node at ~700 W each draws ~5.6 kW; a 1% improvement in mean PAR is a ~1% cut in compute energy, the dominant term. Across a fleet running 24/7, that compounds into real kilowatt-hours and real carbon — which is the argument that turns "we improved a leaderboard ratio" into "we cut the energy bill of serving this model."

The composite energy score is the bridge: it expresses every optimization in the one unit a datacenter operator actually pays for.

What to put on the slide

- The competition scores *time*; we re-expressed it as **energy (joules)**, using the real H100-SXM / NVLink-4.0 figures the simulator already assumes.
 - **1 PAR point $\approx$ 42 kJ; 1 expert move $\approx$ 0.069 J $\rightarrow$ one PAR point is worth $\sim$ 600,000 moves.** Balance is the energy cost; movement is nearly free.
 - Our energy score ranks the versions identically to the grader, but in physical units — and shows compute is 99%+ of consumption.
 - It explains v2's failure as a literal energy waste ($\sim$ 43 kJ), not an abstract point loss.
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Constants used (all reproducible)

python

```
H100_TDP_W          = 700.0          # H100 SXM, NVIDIA datasheet
BALANCED_COMPUTE_S   = 60.0          # simulator
EXPERT_BYTES         = 88_080_384    # simulator
BANDWIDTH_BPS        = 900e9         # NVLink 4.0 (simulator)
NVLINK_PJ_PER_BIT    = 1.3           # NVIDIA
HBM3_PJ_PER_BIT       = 0.29         # HBM3 measurement

E_PAR_POINT_J        = H100_TDP_W * BALANCED_COMPUTE_S          # 42,000 J
E_MOVE_J_tdp         = H100_TDP_W * EXPERT_BYTES / BANDWIDTH_BPS # 0.069 J (conservat
E_MOVE_J_bits        = EXPERT_BYTES*8 * (1.3+0.29)*1e-12         # 0.0011 J (datapath)
```